Ionic Compounds

16. Which of the following substances would you expect to be ionic substances?

a) Substance A is a soft yellow substance that melts at about 40°C.

b) Substance B is a colourless liquid at room temperature.

c) Substance C is a white brittle solid which melts at about 800°C and is soluble in water.

d) Substance D is a brown gas (at room temperature) with an irritating smell.

e) Substance E is a green powder with a melting point of about 900°C. It is not soluble in water.

17. Draw electron arrangement diagrams to show the rearrangement of electrons that occurs when the following atoms react:

a) lithium with chlorine

b) magnesium with oxygen

c) sodium with nitrogen

d) aluminium and fluorine

e) sodium and oxygen

18. The following questions refer to the reaction between calcium and sulfur.

a) Will an ionic compound form in this reaction? Explain how you decided on your answer.

b) Give the electron configurations of calcium and of sulfur.

c) Which atom will lose electrons, and how many, in the reaction?

d) Which atom will gain electrons in the reaction, and how many?

e) How many atoms of sulfur will be required to react with each atom of calcium?

f) Give the formulae of the ions formed (e.g. Na^+) and their electron configurations.

g) Give the formula and the name of the compound formed in this reaction.

h) Give two properties of this compound.

19. The following questions refer to the reaction between aluminium and oxygen.

a) Will an ionic compound form in this reaction? Explain how you decided on your answer.

b) Give the electron configurations of aluminium and of oxygen.

c) Which atom will lose electrons, and how many, in the reaction?

d) Which atom will gain electrons in the reaction, and how many?

e) How many atoms of oxygen will be required to react with every two atoms of aluminium?

f) Give the formulae of the ions formed (e.g. Na^+) and their electron configurations.

g) Give the formula and the name of the compound formed in this reaction.

h) Give two properties of this compound.

20. Complete the gaps in the table below, the first example has been completed:

Element	Number of protons and electrons in neutral atom	Electron arrangement of neutral atom	Number of electrons lost or gained in the formation of an ionic compound	Number of protons and electrons in ion formed	Formula of ion formed	Name of ion formed
calcium	20 protons 20 electrons	2, 8, 8, 2	2 electrons lost	20 protons 18 electrons	Ca^{2+}	calcium ion
potassium						
chlorine						
nitrogen						
					Al^{3+}	
		2, 6				
	12 protons					

21. a) How many electrons are in the outershell of the elements from Group 1 in the Periodic Table?

b) How many electrons will each atom of a Group 1 element lose when they react with a non-metal?

c) What charge (valency) will the ions have that are formed from a Group 1 element when it reacts with a non-metal?

d) How many electrons will each atom of a Group 2 element lose when it reacts with a non-metal?

e) What charge (valency) will the ions have that are formed from a Group 3 element when it reacts with a non-metal?

22. a) How many electrons are in the outershell of the elements from Group 7 in the Periodic Table?

b) How many electrons will each atom of a Group 7 element gain when they react with a metal?

c) What charge (valency) will the ions have that are formed from a Group 7 element when it reacts with a metal?

d) How many electrons will each atom of a Group 6 element gain when it reacts with a metal?

e) What charge (valency) will the ions have that are formed from a Group 5 element when it reacts with a metal?

23. Ionic compounds are formed when metal atoms react with non-metal atoms. Use this information to decide which of the following substances are ionic compounds:

a) iron oxide
b) carbon dioxide
c) water
d) copper chloride
e) potassium sulfide
f) tin
g) $ZnCO_3$
h) SO_3

24. Give the names of the following positive ions:

a) Li^+
b) Ca^{2+}
c) Fe^{3+}
d) Fe^{2+}
e) Ni^{2+}
f) NH_4^+

25. Write the formula of the following positive ions:

a) sodium ion
b) gold ion
c) iron(II) ion
d) aluminium ion
e) lead ion
f) silver ion
g) mercury ion
h) ammonium ion
i) barium ion
j) magnesium ion
k) potassium ion
l) copper ion

26. Write the names of the following negative ions:

a) F^-
b) S^{2-}
c) SO_3^{2-}
d) SO_4^{2-}
e) HSO_3^-
f) HSO_4^-
g) P^{3-}
h) HCO_3^-.

27. Give the formulae of the following negative ions:

a) chloride ion
b) carbonate ion
c) hydroxide ion
d) nitride ion
e) nitrate ion
f) nitrite ion
g) bromide ion
h) phosphate ion
i) phosphide ion
j) sulfate ion
k) hydrogensulfate ion
l) iodide ion

28. Give the names of the following negative ions:

a) S^{2-} SO_3^{2-} SO_4^{2-}

b) NO_3^- NO_2^- N^{3-}

c) P^{3-} PO_4^{3-}

d) O^{2-} OH^-

29. Give the names of the ionic compounds that are made up of the following pairs of ions:

Ions in the ionic compound	**Name of the ionic compound**
calcium ions and sulfate ions	
sulfide ions and iron (III) ions	
Cu^{2+} ions and OH^- ions	
HCO_3^- ions and Al^{3+} ions	

30. Give the names of the ions that make up each of the following ionic compounds, and then name the compound:

Formula of compound	Names of ions in compound	Name of compound
KCl ZnF_2 $NaNO_3$ $Ba(OH)_2$		

31. Draw a small portion of the lattice of ions (showing the ions that are present) that makes up each of the following ionic substances:

a) magnesium oxide b) lead carbonate c) KOH

32. Name the following ionic compounds:

a) AgBr
b) CuO
c) Mg_3P_2
d) Li_2CO_3
e) $Al_2(SO_4)_3$
f) $Fe(HCO_3)_3$
g) NH_4Cl
h) $FePO_4$

33. Give the formula of each of the following ionic compounds:

a) sodium iodide

b) calcium chloride

c) gold bromide

d) silver oxide

e) zinc sulfite

f) aluminium sulfide

g) potassium phosphate

h) copper hydrogensulfate

i) barium nitrate

j) lithium phosphide

k) magnesium nitrite

l) iron(II) hydroxide

34. a) Potassium dichromate is made up of potassium ions and dichromate ions ($Cr_2O_7^{2-}$). What is the formula of potassium dichromate?

b) Potassium permanganate has the formula of $KMnO_4$. What is the formula of the permanganate ion?

c) Calcium chlorite has the formula $Ca(ClO_2)_2$. What is the formula of the chlorite ion?

d) Stannic chloride has the formula of $SnCl_4$. What is the formula of the stannic ion? What is another name for the stannic ion?

35. a) The formula of magnesium hydrogenphosphate is $MgHPO_4$. What is the formula of sodium hydrogenphosphate?

b) Manganese bromide has the formula of $MnBr_2$, and aluminium hypochlorite has the formula of $Al(ClO)_3$. What is the formula of manganese hypochlorite?

36. Strontium (Sr) is a Group 2 element.

a) How many electrons would be in the outermost shell of a strontium atom?

b) Give the formula of the monatomic ion you would expect strontium to form when it reacts with a non-metal.

c) Give the formulae of i) strontium nitrate

ii) strontium phosphate.

5. Molecular Substances

37. Draw electron arrangement diagrams to represent the formation of molecules that occurs when the following atoms join together:

a) two hydrogen atoms

b) hydrogen atom + fluorine atom

c) fluorine atom with a chlorine atom

d) oxygen atom with a sulfur atom to form sulfur monoxide, SO.

38. Draw the electron arrangement diagram to represent the following molecules:

a) a molecule formed between two chlorine atoms (Cl_2)

b) a molecule formed between two hydrogen atoms and one sulfur atom (H_2S) (the sulfur atom is in the middle between the two hydrogen atoms)

c) a molecule formed between one nitrogen atom and three hydrogen atoms (NH_3) (the nitrogen atom is in the middle)

39. Draw the structural formula for each of the molecules formed in questions 37 & 38.

40. Which of the following substances are ionic compounds and which are molecular substances?

a) copper oxide

b) nitrogen dioxide

c) $FeCl_3$

d) sodium sulfate

e) octane, C_8H_{18}

f) fluorine, F_2.

41. Below are descriptions of five substances. Identify each by choosing a name from the substances listed in the box

copper sulfate	carbon dioxide	glucose ($C_6H_{12}O_6$)	sodium chloride	petrol (C_8H_{18})

Substance	**Colour and state at room temperature**	**Melting point (^oC)**	**Solubility in water**
A	White solid	150	Soluble
B	Blue solid	600	Soluble
C	Colourless liquid	- 50	Insoluble
D	White solid	600	Soluble
E	Colourless gas	- 44 (sublimes)	Slightly soluble

42. Choose the correct words:

a) Ammonia (NH_3) is a (*ionic compound / molecular compound*) and it is composed of (*ions / molecules*) in which atoms are bonded together by (*ionic bonds / covalent bonds*). In a molecule of ammonia, the three hydrogen atoms and the nitrogen atom (*share / gain / lose*) electrons, and (*ionic / covalent*) bonds exists between the hydrogen and nitrogen atoms. Ammonia would probably be a substance with a (*low / high*) melting point.

b) Sodium oxide is an (*ionic compound / molecular compound*) and it is composed of (*ions / molecules*) which are bonded together by (*ionic bonds / covalent bonds*). In the formation of sodium oxide, the sodium atoms (*gained/lost/shared*) electrons and the oxygen atoms (*gained/lost/shared*) electrons. Sodium oxide is composed of (*molecules/ positive and negative ions*) and it has a formula of (*NaO / NaO_2 / Na_2O*). Sodium oxide would probably be a (*solid / liquid / gas*) at room temperature.

43. a) What is the formula of the molecules of which liquid water is composed?

b) What is the formula of the molecules that make up ice?

c) When ice melts, what bonds are broken?

d) Use your answer to c) to explain why ice has a low melting point.

e) When sodium chloride melts, the positive sodium ions move apart from the negative chloride ions. Would you expect the ionic bonds between the ions in sodium chloride to be stronger or weaker than the bonds between the molecules in ice?

44. Write the formulae of the following molecular substances:

Number and type of atoms in each molecule of the substance	Formula of the substance
one sulfur atom and two oxygen atoms	
three carbon atoms and eight hydrogen atoms	
six carbon atoms. twelve hydrogen atoms and six oxygen atoms	

45. Write the formula for each of the following molecular compounds:

a) nitrogen monoxide
b) sulfur dioxide
c) diphosphorus pentoxide
d) silicon tetrachloride
e) dinitrogen monoxide
f) sulfur hexachloride

46. Write the name for each of the following molecular compounds:

a) SO_3
b) CO
c) PBr_3
d) Si_2Cl_6

47. Identify each of the following substances as an element or a molecular compound or an ionic compound

a) water
b) oxygen
c) helium
d) carbon dioxide
e) methane (CH_4)
f) copper
g) sodium chloride
h) ammonia (NH_3)

6. Metals & Covalent Network Substances

48. Identify each of the following substances as a metal, an ionic substance or a molecular substance.

a) iron
b) water
c) copper
d) carbon dioxide
e) sodium chloride
f) oxygen
g) copper sulfate
h) hydrogen fluoride
i) calcium nitrate

49. Below are given some descriptions of eight substances. Identify each substance by choosing a name from the following group

copper, oxygen, copper sulfate, sugar ($C_{12}H_{22}O_{11}$), aluminium, water, sodium chloride, diamond

a) a colourless liquid at room temperature
b) a colourless gas which can be found in the air
c) a shiny yellow, brown-coloured solid which is a very good conductor of electricity
d) a colourless solid which is very hard and has a very high melting point
e) a white solid that has a high melting point
f) a blue solid that is soluble in water
g) a white solid with a low melting point
h) a shiny grey solid which conducts electricity

50. The metal M forms an ion with the formula M^{2+}. What is the formula of the chloride of M and the sulfate of M?

51. The element X reacts with chlorine to form the ionic compound XCl. What is the formula of the oxide of X?

52. The element A forms the ion A^{2-}. What is a possible formula of the ionic compound formed when A and gold react?

53. The element D reacts with zinc to form the ionic compound ZnD_2. What is a possible formula of the ionic compound formed when D reacts with potassium?

54. From the following list of substances, choose your answers to the questions below:

water (H_2O), sodium fluoride (NaF), chlorine monofluoride (ClF), sulfur monoxide (SO),
copper (Cu), calcium phosphide (Ca_3P_2), silicon dioxide (SiO_2), silane (SiH_4)

a) Which two substances are ionic compounds?
b) Which substance is an element?
c) Which solid substance would you expect to be a very good conductor of electricity?
d) Which substance is a covalent network compound?
e) Name one substance which is composed of positive and negative ions.
f) Which four substances which are composed of molecules?
g) In which substance are the atoms in the molecules bonded together by a double covalent bond?
h) Name two substances you would expect to have low melting points.
i) In which two substances are the atoms held together by ionic bonds.
j) In which substance are the atoms arranged in molecules in which there are four single covalent bonds?
k) In which five substances do the atoms share electrons?